Dr. Manjunatha C

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1. Professional Positioning

I am a technology strategist and applied research expert with over 20 years of experience in translating advanced nanomaterials and electrochemistry into **scalable, industry-ready solutions**. As Head of the Centre of Excellence in Nanomaterials and Devices (CND) at RV College of Engineering, Bengaluru, I specialize in bridging **lab-to-market innovation** across energy storage (Li-ion batteries, supercapacitors), green hydrogen, sensors, and industrial nanomaterials.

With established expertise in **500+ nanomaterial systems**, 120+ publications, and patented technologies, I work closely with industry partners in EVs, clean energy, coatings, catalysts, and biomedical sectors to **accelerate R&D, optimize prototypes, and enable commercialization**.

2. Core Technical Expertise

- Design, engineering, and scale-up of high-performance electrode materials for lithium-ion batteries and asymmetric supercapacitors
- Development of transition metal oxides, chalcogenides, and hybrid carbon–inorganic nanocomposites for advanced energy storage and conversion systems
- Synthesis and optimization of catalytic nanomaterials for green hydrogen production (HER/OER applications)
- Engineering of high-sensitivity electrochemical and gas sensors for biomedical diagnostics (e.g., glucose, uric acid, vitamin C) and environmental monitoring (heavy metals, toxic ions, and gases)
- Electrochemical modeling, simulation, and digital twin development for performance prediction and lifecycle optimization of lithium-ion batteries in electric vehicles
- Technical expertise in the design, customization, and supply of functional nanomaterials for industrial applications, including lubricants, ceramic pigments, paints, coatings, nanocomposites, and catalytic systems
- Development of application-specific nanomaterials tailored for performance enhancement such as wear resistance, thermal stability, corrosion protection, conductivity, and catalytic efficiency

3. Industry Problems I Can Solve

- **Scalable Nanomaterial Manufacturing:** Develop cost-effective, scalable synthesis routes for advanced nanomaterials tailored to industries including battery/EV technology, coatings, paints, ceramic pigments, nanolubricants, composites, and textiles

- **Materials Characterization & Failure Analysis:** Provide in-depth structure–property correlation and root-cause analysis using advanced techniques (SEM, TEM, XRD, Raman, FTIR, UV-Vis, BET, XPS, X-ray CT, TGA-DSC) for performance optimization in semiconductors, catalysts, and functional materials

- **Battery & EV Performance Optimization:** Enhance energy density, cycle life, and safety of lithium-ion batteries and supercapacitors through advanced electrode design, degradation analysis, and electrochemical diagnostics

- **Electrode Degradation Mitigation:** Develop strategies to minimize capacity fading and structural instability during long-term charge–discharge cycling in energy storage systems

- **Prototype Development & Device Engineering:** Optimize asymmetric supercapacitor coin cells and electrode architectures for wearable electronics, portable devices, and EV applications

- **Digital Twin & Predictive Modeling:** Accelerate R&D cycles through electrochemical modeling and digital twin frameworks for battery systems, enabling performance prediction and lifecycle optimization

- **Advanced Electrochemical Diagnostics:** Diagnose system inefficiencies and failures using EIS, CV, GCD, and DPV for batteries, catalysts, and sensor platforms

- **Hydrogen & Catalyst Engineering:** Improve electrocatalyst efficiency for hydrogen and oxygen evolution reactions (HER/OER) for green hydrogen technologies and industrial catalytic processes

- **Sensor Technologies (Environmental & Biomedical):**
  - Develop highly selective gas sensors (e.g., CO_2, SO_2) for environmental monitoring, cleanrooms, and industrial safety
  - Design non-enzymatic electrochemical biosensors for clinical diagnostics (glucose, uric acid, vitamin C)

- **Industrial Nanomaterials Solutions:** Provide application-driven nanomaterials for:
  - **Coatings & Paints:** corrosion resistance, thermal stability, functional coatings
  - **Ceramic Pigments:** color stability, high-temperature performance
  - **Lubricants:** friction reduction, wear resistance, thermal durability
  - **Composites & Textiles:** mechanical strength, conductivity, multifunctionality

- **Graphene & Advanced Nanocomposites:** Develop graphene-based and hybrid nanocomposites for electronics, EMI shielding, energy devices, and structural applications

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- **Semiconductor & Functional Materials Support:** Assist in material design, defect analysis, and performance enhancement for semiconductor and electronic material systems

- **Defence & Strategic Applications:** Develop high-performance nanomaterials for sensing, energy systems, protective coatings, and advanced composites

- **Agricultural Nanotechnology:** Design nanomaterials for smart fertilizers, sensing, and environmental monitoring to improve agricultural efficiency and sustainability

4. Key Consultancy / Project Offerings

- **Nanomaterial Design, Synthesis & Scale-up:**
  Custom development and scalable production of advanced inorganic nanomaterials tailored for batteries, hydrogen technologies, catalysts, coatings, paints, ceramic pigments, nanolubricants, composites, textiles, agriculture, and defence applications

- **Battery & Supercapacitor Engineering (EV Focus):**
  End-to-end development of electrode materials, coin cell fabrication, performance testing, and optimization for lithium-ion batteries and supercapacitors targeting EV and portable energy storage applications

- **Electrochemical System Diagnostics & Failure Analysis:**
  Root-cause analysis of performance degradation using advanced electrochemical techniques (EIS, CV, GCD, DPV) and material characterization to improve reliability, safety, and lifecycle

- **Hydrogen & Catalytic Systems Development:**
  Design and optimization of high-performance electrocatalysts for hydrogen and oxygen evolution reactions (HER/OER) and industrial catalytic processes

- **Sensor & Device Development (Biomedical & Environmental):**
  End-to-end design and prototyping of electrochemical sensors, gas sensors, and biosensors for healthcare diagnostics, environmental monitoring, and industrial safety applications

- **Digital Twin & Predictive Modeling for Energy Systems:**
  Development of simulation-driven frameworks for performance prediction, degradation analysis, and lifecycle optimization of battery systems in EVs and grid storage

- **Industrial Nanomaterials Solutions & Product Integration:**
  Application-specific material design and deployment for:

- Coatings, paints, and corrosion-resistant systems
- Ceramic pigments and high-temperature materials
- Nanolubricants for friction and wear reduction
- Graphene and nanocomposites for EMI shielding and structural applications

- **Materials Characterization, Validation & Optimization:**
  Comprehensive structural, morphological, and functional analysis (SEM, TEM, XRD, Raman, FTIR, BET, XPS, etc.) to guide product development, quality control, and performance enhancement

- **Semiconductor & Advanced Materials Support:**
  Material design, defect analysis, and process optimization for electronic, semiconductor, and functional material applications6. Selected High-Impact Work

5. Technical Capabilities, Infrastructure & Proven Impact

- **Advanced Nanomaterials Development & Scale Expertise:**
  Established synthesis protocols for over **500+ nanomaterials systems**, supported by **1000+ experimental studies**, enabling reliable translation from lab-scale innovation to industry-relevant applications

- **High-Impact Technology Development:**
  - Developed **patented Ni-based nanocomposites** and **tungsten oxide nanocubes**, achieving enhanced energy density in asymmetric supercapacitors
  - Fabricated **high-performance coin cell prototypes** using carbon nanotube-based nanocomposites for next-generation energy storage systems
  - Engineered **advanced electrocatalysts** based on transition metal oxides and chalcogenides for improved green hydrogen generation (HER/OER)
  - Designed **highly selective, non-enzymatic electrochemical biosensors** for accurate detection of uric acid and vitamin C
  - Developed **low-trace gas sensors** for SO_2 and CO_2 monitoring in environmental and cleanroom conditions

- **World-Class Research & Laboratory Ecosystem:**
  Access to advanced **national and international laboratories** through established research collaborations, enabling cutting-edge material synthesis, characterization, and device validation

- **Global Research & Industry Network:**
  Strong collaborations with **internationally acclaimed Top 2% scientists**, along with a well-established network across academia, national laboratories, and industry sectors including energy, materials, and electronics
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- **Industrial & Academic Collaboration Strength:**
  Proven track record of working with industry partners for applied R&D, technology transfer, and product-oriented development in nanomaterials and electrochemical systems

- **Publications, Patents & Scientific Contribution:**
  - **120+ research publications** in peer-reviewed Scopus/Web of Science indexed journals

  - **6 Indian patents** focused on scalable and application-oriented technologies

- **Talent Development & Technical Leadership:**
  Supervised **150+ undergraduate and postgraduate engineering projects**, building a strong pipeline of skilled researchers and engineers

- **Global Alumni & Talent Network:**
  Established a strong alumni network with former students working as **research scientists and engineers** in leading companies, universities, and national laboratories in India and abroad

- **Professional Visibility & Outreach:**
  Active professional presence with a **strong LinkedIn network (7,000+ connections and 8,000+ followers)**, enabling industry engagement, knowledge dissemination, and collaboration opportunities

7. Differentiation / Unique Value Proposition

Recognized among the Top 2% Scientists Globally (2023), I combine over 20 years of expertise in nanomaterials and electrochemistry with a strong focus on scalable, industry-ready solutions. With 120+ publications, 6 patents, and established synthesis protocols for 500+ nanomaterials, I specialize in translating advanced research into practical technologies across energy storage, hydrogen, sensors, coatings, and catalysts.

My core strength lies in bridging lab-to-market innovation—from material design and structure–property optimization to device prototyping and digital twin modeling. Supported by a robust network of international collaborators, advanced laboratories, and industry partnerships, I deliver end-to-end solutions that are scientifically sound, commercially viable, and rapidly deployable.

8. Engagement Models

- Technology Consulting: Strategic advisory for R&D scaling and material selection.
- Sponsored R&D: Focused project execution for custom component development.
- Joint Development Programs: Collaborative partnerships for next-gen technology.
- Technology Transfer: Licensing and transferring patented IP to industrial manufacturing.

9. Contact & Collaboration Note

I actively seek strategic collaborations with industry leaders, R&D organizations, and investors working in energy storage, hydrogen technologies, advanced materials, sensors, and emerging nanotechnology applications. I welcome opportunities to co-develop scalable solutions, accelerate product innovation, and translate cutting-edge research into commercially viable technologies.

Let us connect to explore how advanced nanomaterials and electrochemical innovations can deliver **measurable performance improvements, cost efficiency, and sustainable business impact**.

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